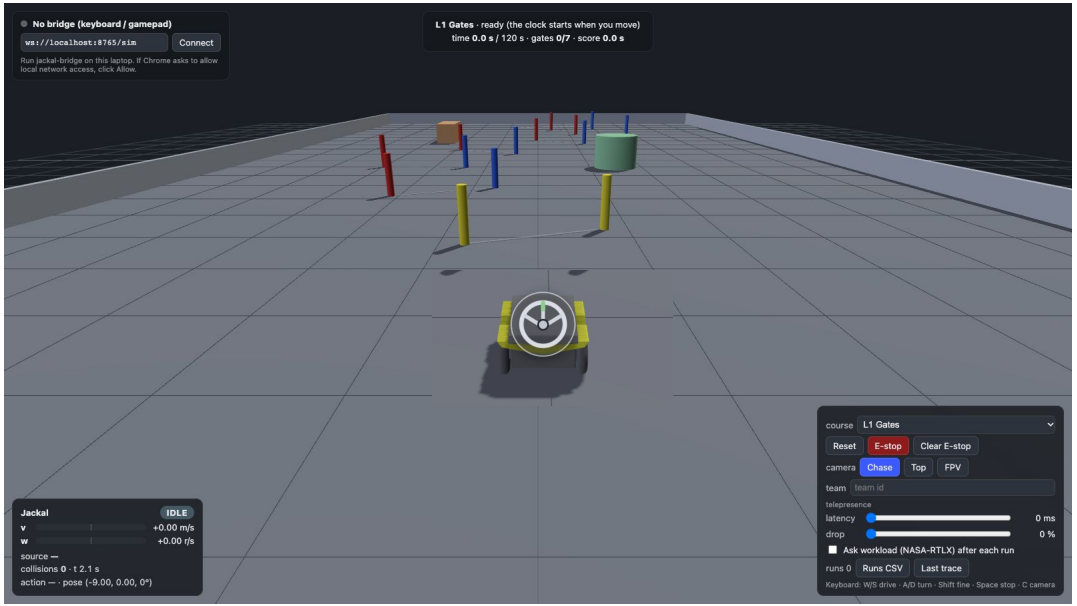
Myo-Controlled Jackal

Nicholas, Orb, Shima and Ester

Muscles set speed
Arm motion steers
LiDAR limits speed

Calibration + fixed rules
my_team3_2.py



Simulator reference view

Earlier Prototype: Trained Classifier

- **Four action classes**: forward, backward, left, right
- **42 recordings** from one participant
- **Eight-channel RMS**, scaling and shrinkage LDA
- **Five folds** grouped by recording
- **Final V3.2:** classifier replaced by **calibration + rules**



Real Myo signal check
Earlier hardware verification

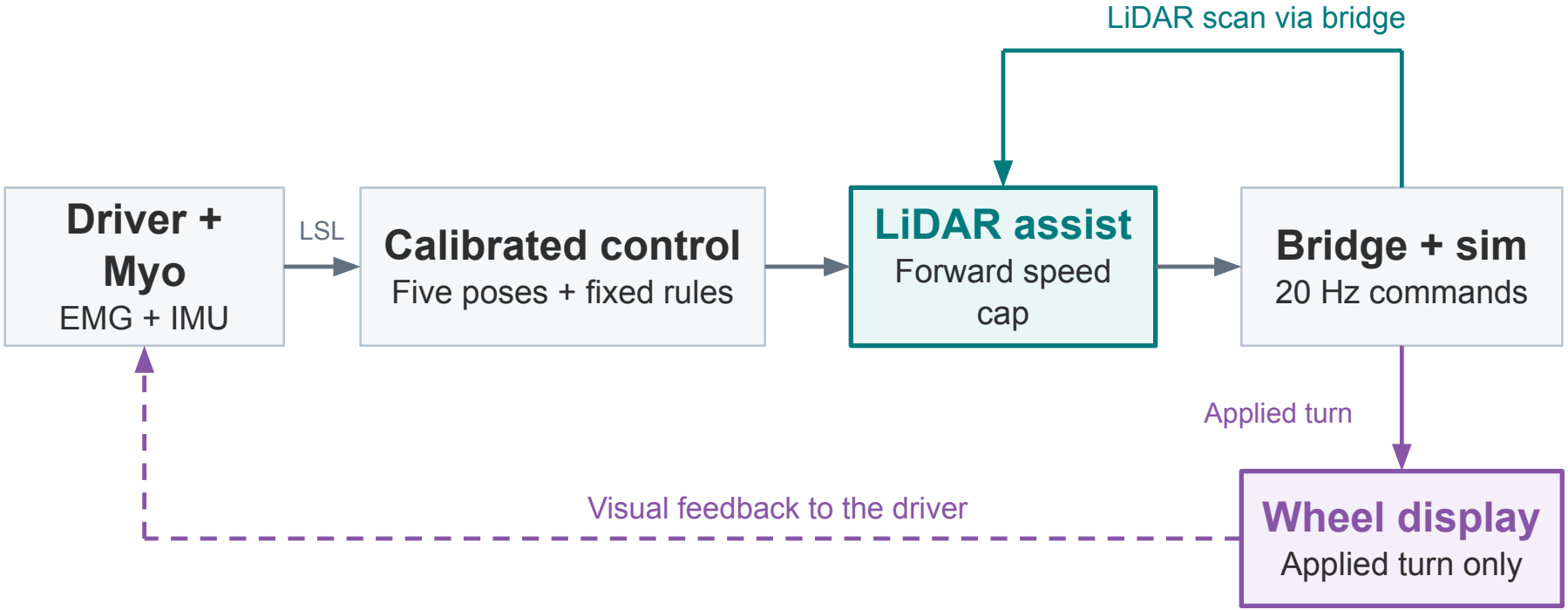
Our Contribution: Myo Control + LiDAR

- **Speed:** eight-channel EMG effort scales forward speed
- **Direction:** IMU twist steers, arm lift requests reverse
- **Personalization:** five calibration poses set the ranges
- **Assistance:** LiDAR caps speed, the driver steers
- **Feedback:** wheel shows the applied turn command



Real Myo IMU check
Orientation and motion channels

System Architecture:
LiDAR + Wheel Feedback (Signifier)



Personal calibration + fixed rules. No ML classifier training.

How the Driver Controls V3.2



Relax	Fist	Lift forearm
0.4 m/s forward	**Up to 2.0 m/s**	**0.6 m/s reverse**

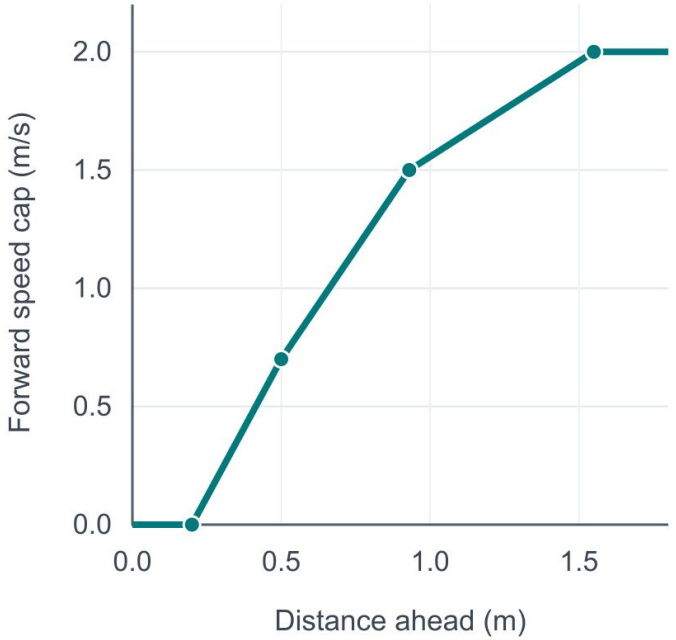
- **LiDAR table** sets the forward-speed ceiling
- Cap drops immediately, recovers up to **0.5 m/s²**
- **20 Hz** commands, **dropout stops** and a zero tick on reversal
- **Deliberate stop:** simulator **E-stop** or **Ctrl+C**

V3.2: LiDAR Distance-to-Speed Limits

Distance (m)	Speed cap (m/s)
0.20	0.00 (stop)
0.50	0.70
0.93	1.50
1.55	2.00

- **Linear interpolation** between points
- Cap recovery: at most **0.5 m/s²**
- Cap reductions: **immediate**

LIDAR CAPS SPEED, DRIVER STEERS



Target curve before recovery smoothing

Final Method: Calibration + Fixed Rules

- Final V3.2: **no classifier training or loading**
- **Five poses:** relax, left/right twist, lift and fist
- **EMG:** relaxed and fist levels scale muscle effort
- **IMU:** neutral and arm poses define directions
- **Fixed rules:** equations, thresholds and smoothing
- **~30 s** per attempt, with quality checks and retry



Armband placement illustration

Hold the forearm in the air for calibration

L1–L3 Results

Displayed finished scores (0 ms latency, 0% loss)

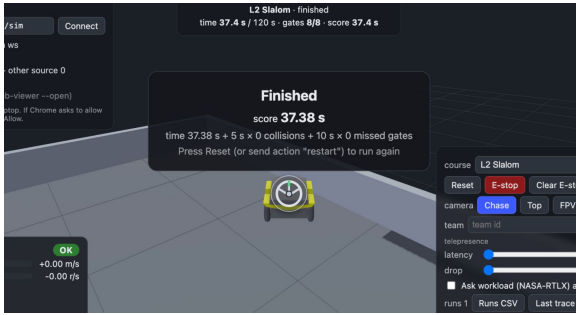
L1 Gates

12.38 s



L2 Slalom

37.38 s



L3 Corridor / dock

18.66 s



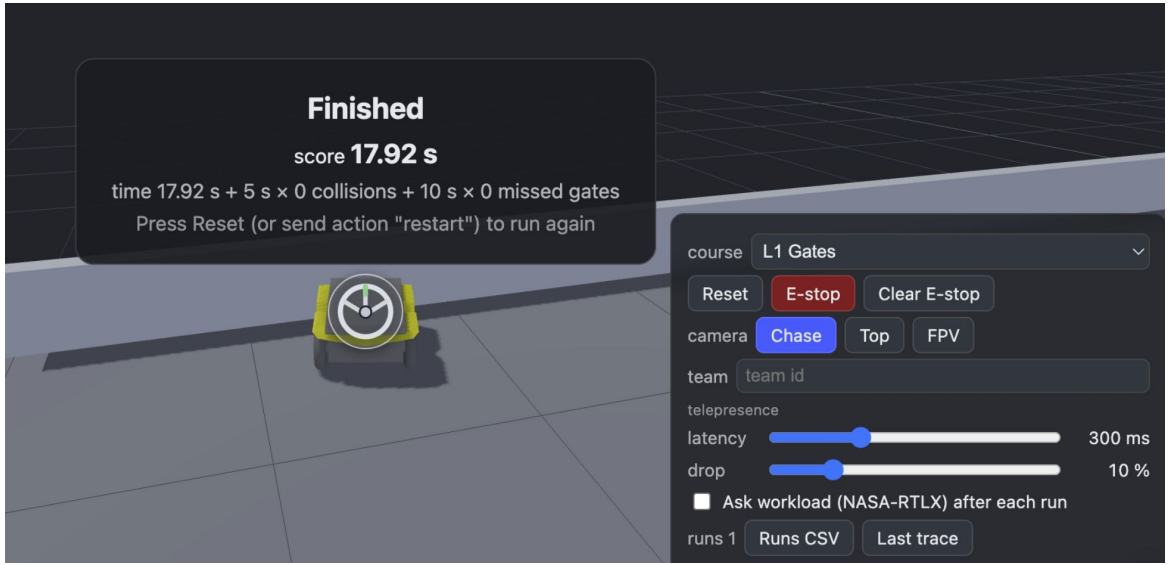
Score = time + **5 s per collision** + **10 s per missed gate**

L1 Noisy Results

Displayed finished scores (300 ms latency, 10% loss)

L1 Gates

17.92 s



Score = time + **5 s per collision** + **10 s per missed gate**

Future Research: Keeping Users in Control

- **Learning:** how do novices master body-based control?
- **Awareness:** does the wheel clarify robot responses?
- **Agency:** when does LiDAR help or feel restrictive?
- **Trust:** can users predict when assistance intervenes?
- **Adaptation:** how does refitting the band affect control?

IMPLEMENTED: TURN FEEDBACK



Next: evaluate awareness and sense of control